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# Solar Eligibility Token (SET)

A Blockchain-Enabled Marketplace Framework to Standardize and Scale RTS Deployment Nationwide


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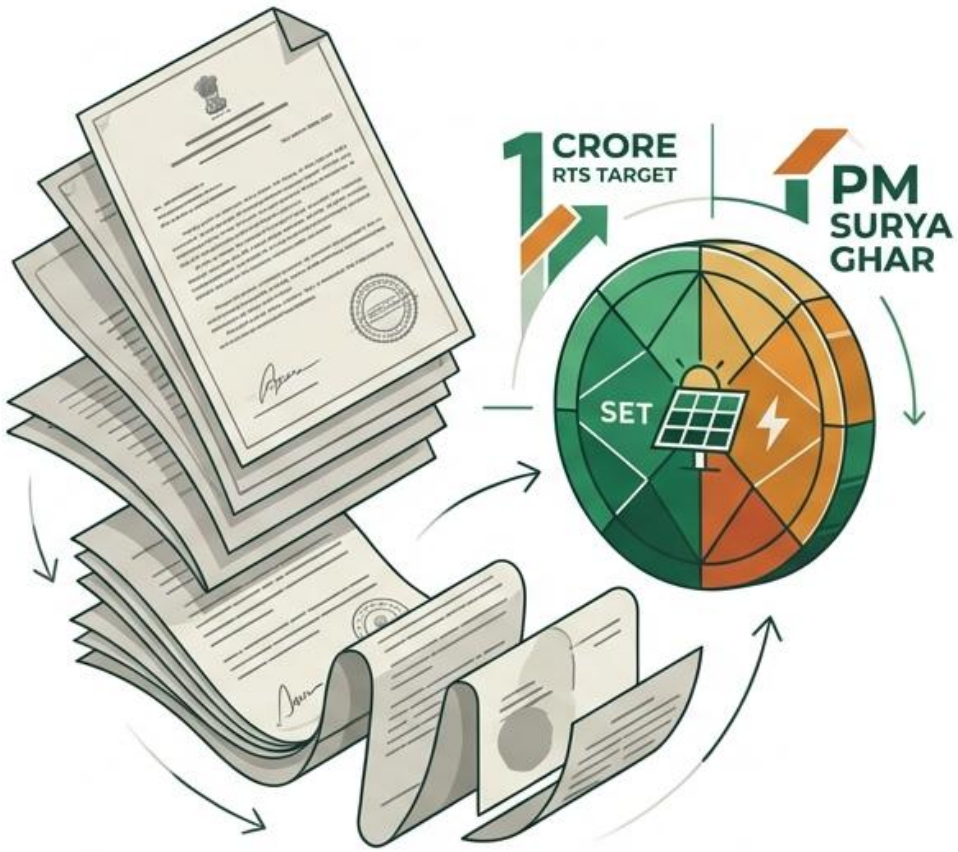
# INTRODUCTION

**The National RTS Challenge:** Rooftop Solar (RTS) is critical for the renewable energy transition, but India's market expansion is heavily fragmented and slowed by inconsistent state policies.

**The Solution:** The Solar Eligibility Token (SET) is a blockchain-enabled strategic digitalization initiative designed to transform bureaucratic bottlenecks into a scalable market instrument.

**Standardizing Adoption:** By moving from tedious manual compliance checks to a transparent, tokenized marketplace, the SET functions as a single unit of value to universally democratize access to solar deployment nationwide.



# Problem Statement

## Identifying eligible customers is currently a stalled, State specific, Manual Process.

Despite successful submission on the PM Surya Ghar Portal, applications often experience extended timelines across the multi-stakeholder approval and verification process, impacting consumer momentum and overall project progress.

**1. Technical Feasibility & Grid Capacity:** Utilities struggle to verify local Distribution Transformer (DT) headroom due to manual grid data.

**2. Connected Load Discrepancies:** Applications exceeding sanctioned load require mandatory enhancements, extending timelines over multiple billing Cycles.

**3. Document Complexity:** Verifying legal roof ownership and structural suitability requires complex cross-checks against disparate land and municipal records.

**3. Financial Vetting and Regulations:** Manual cross-refrencing for pending Discom arrears delays approvals, while diverse SERC criteria hinder Unified National Deployment.

# How SET Token Works ?

Automated trust-less verification via digital layers and multi stockholder token

Applicant Energy Bill

**1. Identity, Ownership & Consent:**
Matches bill against DISCOM CIS database and verifies customer authenticity & premises confirmation.

**2. AI + GIS Service:**
Analyzes rooftop polygons, satellite shading and mathematically confirm shadow free area

**3. Billing/CIS Service:**
Instantly verifies connected load adequacy and absence of pending Discom Arrears.

**4. AMI / SCADA Service:**
Queries Discom Meter database for Consumer ID and last 24 hour load to calculate available headroom.

**SET Token**

**SET Token Marketplace**

**Financers**

**Manufacturers**

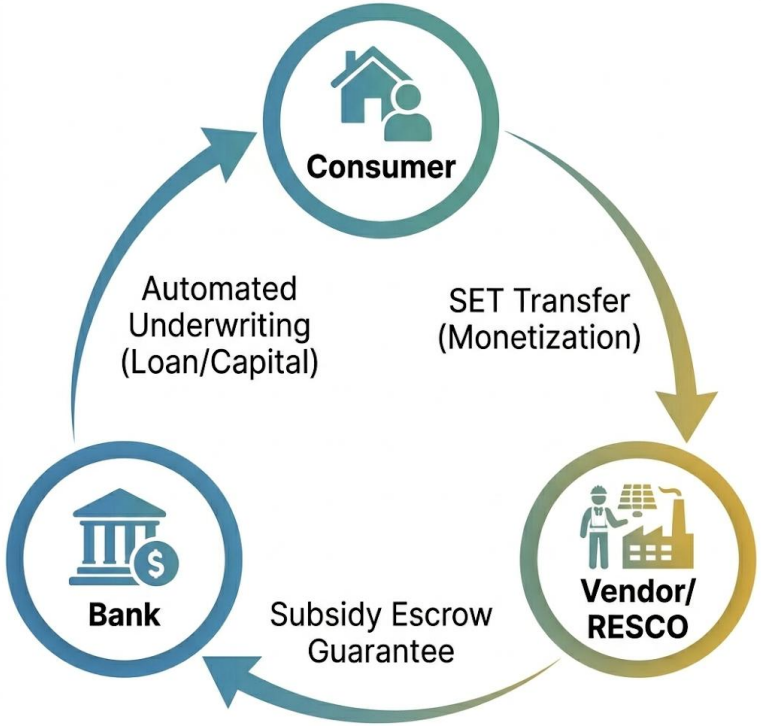
**Discoms**

**RTS Vendors**

# Use case of the SET Token

**Customer Capitalization**
(Rent a Roof)

The customer can transfer their SET (the right to install) to a specialized solar aggregator (RESCO model). This transfer on the blockchain serves as the formal, irrevocable, digital consent for a Power Purchase Agreement (PPA) or lease, instantly monetizing the roof space without any upfront cost to the owner.




**Banks & Escrow**

Bank use the SET as collateral to automate loan underwriting, because the Identity & Technical Clearance is already cryptographically certified.

**Subsidy-Linked Escrow**

Automatically place the subsidy component of the loan into an escrow account linked to the SET's smart contract, guaranteeing payment to the vendor upon final validation and reducing the bank's capital exposure.

# Use case of the SET Token

## Gov & Public Sector

- **Digital Clearances:** Automated municipal building NOCs based on token metadata.
- **Fraud Prevention:** Subsidy releases triggered via smart contracts on Discom validation.
- **Incentive Alignment:** Performance-linked incentives for jurisdictions driving digital adoption.

## Banking & Financial

- **Automated Underwriting:** Compressed loan approvals (days vs weeks) using SET as collateral.
- **Subsidy Escrow:** Smart-contract linked accounts guarantee vendor payment upon project commissioning.
- **Risk Reduction:** Cryptographic verification of subsidy value minimizes credit exposure.

## Vendor & Manufacturer

- **Risk-Free Deployment:** Crew scheduling based on verified SET possession to avoid project rejections.
- **Targeted Sales:** Exclusive discounts for SET holders to ensure high-conversion pipelines.
- **Inventory Optimization:** Precise demand forecasting based on real-time token minting data.

## Customer & Aggregator

- **District Aggregation:** Digital pooling of localized SETs for high-quality bulk procurement.
- **Attribute Monetization:** Pre-selling environmental attributes (Carbon Offsets) for upfront cash.
- **Lower Costs:** Group installation models facilitated by aggregated tokenized eligibility.

## Advanced Grid-Level

- **Virtual Power Plants (VPP):** distributed asset participation in demand response markets.
- **Automated Insurance:** claims processed via tokenized performance metrics and service logs.
- **ESG Reporting:** Verified Scope 2 emission reductions for enterprise corporate accounting.

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# *THANK YOU*

For discussions/suggestions/queries email: isuw@isuw.in

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Links/References (If any)